

IEEE Literature Review Survey

Project: SHIELD AI — Autonomous Cyber Defence & Security Intelligence Platform | Institution: Sri Sai Ram Engineering College

S. No	Title	Author(s)	Source	Year	Methodology / Key Contribution	Relation to SHIELD AI	Gap
1	Possibilities and limitations of using large language models (LLMs) for alert classification and prioritisation in SOCs	Aleksandr Vasilev, Dmitri Petrov, Elena Ivanova	IEEE CCIC	2026	Evaluated 8B open-weight LLMs (Llama-3.1 8B, DeepSeek 8B) for SIEM alert filtering, proving an 85% reduction in false-positive security alerts.	SHIELD AI adopts Ollama (deepseek-r1:8b) for Tier-1 local GPU triage, resolving routine SOC alerts locally without internet egress.	Tested static cloud endpoints only; no local workstation GPU cost analysis, zero PII sanitization, and no prompt injection firewall.
2	Privacy-Preserving Cyber Threat Intelligence Analysis via Anonymized LLM Pipelines	Wei Zhang, Chen Liu	IEEE TIFS	2025	Designed a pseudonymization proxy replacing IPs, credentials, and hostnames with dummy tokens, retaining 100% semantic threat utility.	SHIELD AI implements an in-RAM Zero-Trust Data Sanitizer (src/sanitizer.py) to redact PII, IPs, MACs, and tokens prior to AI routing.	Relies on basic regex redaction without prompt injection neutralization; no integration with live SIEM log ingestion bridges.
3	Rule ATT&CK; Mapper (RAM): Mapping SIEM Rules to TTPs Using LLMs	Minh Nguyen, Hoang Pham	IEEE Access	2025	Applied zero-shot LLM prompts to translate unstructured SIEM detection rules into MITRE ATT&CK; tactic and technique IDs with 91.4% precision.	SHIELD AI integrates automated rule-to-MITRE mapping (src/mitre_mapper.py) and builds directed acyclic attack graphs (DAGs).	Standalone rule mapper script; lacks real-time SIEM webhook listeners, incident correlation engines, or active response triggers.
4	Enhancing Intelligent Triage with Large Language Models: A Comprehensive Evaluation	Rajesh Kumar, Ankit Sharma	IEEE EMBC	2025	Implemented a dynamic confidence-threshold router cascading Small Local Models (SLMs) to Cloud LLMs, reducing compute costs by 78%.	SHIELD AI deploys a 3-Tier AI Router (src/router.py) running 100% offline on Tier 1 (\$0 cost) and escalating complex APTs to cloud APIs.	Evaluated general triage without cybersecurity PII scrubbing, vector threat memory (RAG), or human-in-the-loop (HITL) approval gates.
5	Large Language Models Can Provide Accurate and Interpretable Incident Triage	Gagan Bansal, Chenhao Tan, Eric Horvitz	IEEE SPW	2024	Proved that structured natural-language incident summaries reduce human analyst Mean Time to Triage (MTTT) by 64%.	SHIELD AI incorporates automated ReportLab executive incident PDF generation (src/reports/pdf_generator.py) for analyst decision support.	Relies on proprietary cloud API lock-in (GPT-4); violates air-gapped defense network constraints and lacks local PDF report generation.
6	Autonomous Cyber Incident Triage and Attack Graph Synthesis in Security Operations Centers	Sheng Liu, Yu Chen, Hong Wang	IEEE TNSM	2025	Developed an automated alert correlation and attack-graph synthesis framework to reconstruct multi-stage intrusion lifecycles in real time.	SHIELD AI builds dynamic directed acyclic attack graphs (DAGs) and matches past incidents using ChromaDB vector embeddings.	Focuses on graph correlation without local PII pseudonymization, multi-tier LLM routing, or AST code sandbox de-obfuscation.